AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM240
Introduction to Computer Graphics

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15 April 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 90

Instructions:

- Total Marks: **60**
- This exam consists of **2** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part -I: Question 1: Choose the best correct answer of the following: (15 Marks)

1 The function quadTo has .....parameters.

A 4 B 0
C 3 D 2

2 The function LineTo has .....parameters.

A 3 B 4
C 2 D 1

3 .....space is attached to a device.

A Object B Device
C view D None of the above

4 Combining several existing shapes to build more complicated shape is known as .....

A Constructive area B Winding rules
C Compositing D None of the above

5 ..... is used to define visible colors with numerical values

A Color space B Alpha channel
C Rendering D None of the above

6 The function draw which is a function in the graphics2D object is an example of .....functions

A Modeling B Rendering
C Transformation D None of the above

7 The function drawOval is an example of ..... functions

A Rendering B Modeling
C Modeling and rendering D None of the above

8 The center of the rectangle defined by RoundRectangle2D.Double(a1,a2,a3,a4,a5,a6) is at .....

A (a1+0.5a3,a2+0.5a4) B (a3,a4)
C (a1+0.5a5,a2+0.5a6) D (a5,a6)

9 In the instruction QuadCurve2D.Double(x1, y1, x2, y2, x3,y3) the point controlling trend is .....

A (x1,y1) B (x2,y2)
C (x3,y3) D None of the above

10 .....is a method to associate geometric points with algebraic quantities of ordered tuples of numbers.

A Alpha channel B Coordinate system
C Color system D None of the above

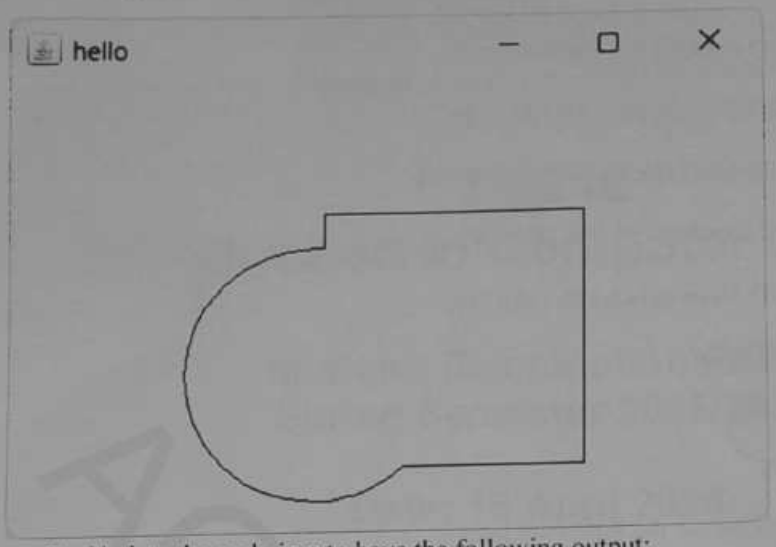
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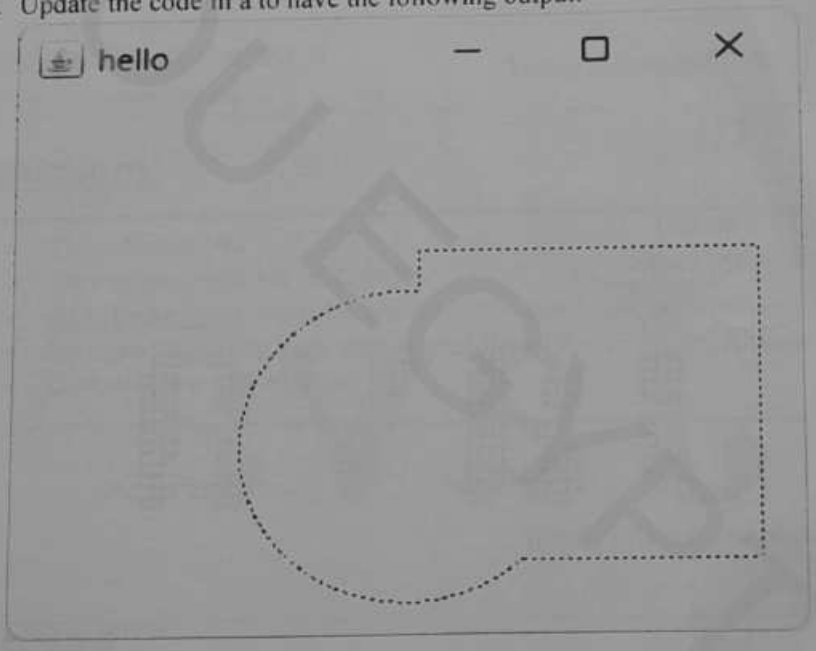
Part-II: Answer all of the following questions (45 Marks).

Question 2: (15 marks)

A. Using constructive area geometry, write a java code segment to draw the following shape:



B. Update the code in a to have the following output:



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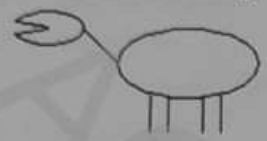
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Question 3: (15 marks)

Having the following code in Java:

```
Shape neck=new Line2D.Double(100,131,81,112);
Shape head=new Arc2D.Double(42,102,39,19,200,320,Arc2D.PIE);
g.draw(neck);
g.draw(head);
Shape lArm=new Line2D.Double(119,150,119,170);
g.draw(lArm);
Shape rArm=new Line2D.Double(129,150,129,170);
g.draw(rArm);
Ellipse2D body=new Ellipse2D.Double(100,112,78,39);
g.draw(body);
Shape Rleg=new Line2D.Double(158,150,158,170);
g.draw(Rleg);
Shape LLeg=new Line2D.Double(148,150,148,170);
g.draw(LLeg);
```

The code has the following out:



Modify the above code to output the following image:



Question 4: (15 marks)

Having the following shape:



Write a java code segment to plot it.

End of questions

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